

DSC 245 Project — Section 2.1

Macroeconomic Indicators & Stock Market: A Causal Analysis

Team 2.1

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Contents

1	Data	2
2	Exploratory Analysis	3
2.1	Time Series	3
2.2	Correlation Matrix	4
2.3	Rolling Correlation	5
3	Methods & Results	5
3.1	Granger Causality	5
3.2	PC Algorithm	7
3.3	CD-NOD (Nonstationary Causal Discovery)	9
4	Causal Effect Estimation	10
4.1	Effects on S&P 500 Return	10
4.2	Effects on Fed Funds Rate Δ	11
5	Summary of Findings	12

Data

We collected **288 months of data (January 2001 – December 2024)** from two sources:

- **Yahoo Finance:** S&P 500 ($\sim\text{GSPC}$), VIX ($\sim\text{VIX}$)
- **FRED:** Federal Funds Rate, CPI, Unemployment Rate, M2 Money Supply, Consumer Sentiment, WTI Oil Price, 10-Year Treasury Yield

All variables were transformed to ensure stationarity before modeling:

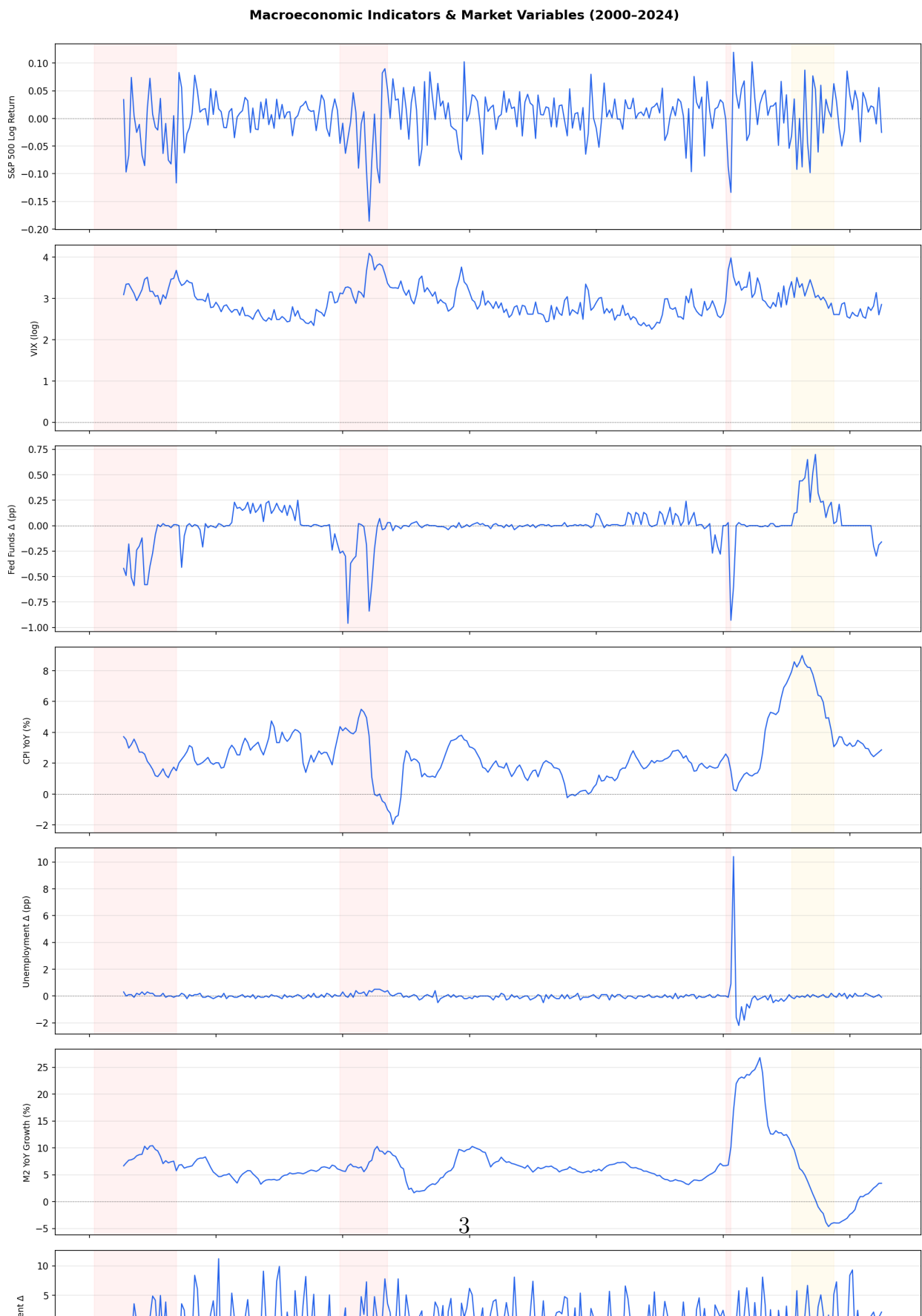
Table 1: Variable Transformations

Variable	Transformation	Rationale
S&P 500	Monthly log return	Price level is non-stationary
VIX	Log level	Already mean-reverting
Fed Funds Rate	First difference (MoM)	Level is non-stationary
CPI	Year-over-year % change	Captures inflation rate
Unemployment	First difference (MoM)	Level is non-stationary
M2	Year-over-year % change	Captures growth rate
Consumer Sentiment	First difference (MoM)	Level is non-stationary
Oil	Monthly log return	Price level is non-stationary
10Y Treasury	First difference (MoM)	Level is non-stationary

All 9 transformed variables passed the **ADF stationarity test** at $\alpha = 0.05$ (CPI YoY $p = 0.043$, all others $p < 0.01$).

Exploratory Analysis

Time Series



Three key episodes stand out:

- **2008–2009 GFC:** Fed Funds Rate dropped nearly 1pp in a single month; unemployment spiked sharply.
- **2020 COVID:** Unprecedented M2 growth ($\sim 25\%$ YoY); unemployment shock of +9pp in one month.
- **2022–2023 Rate Hike Cycle:** CPI peaked at $\sim 9\%$ YoY; Fed raised rates by up to 0.75pp/month.

Correlation Matrix

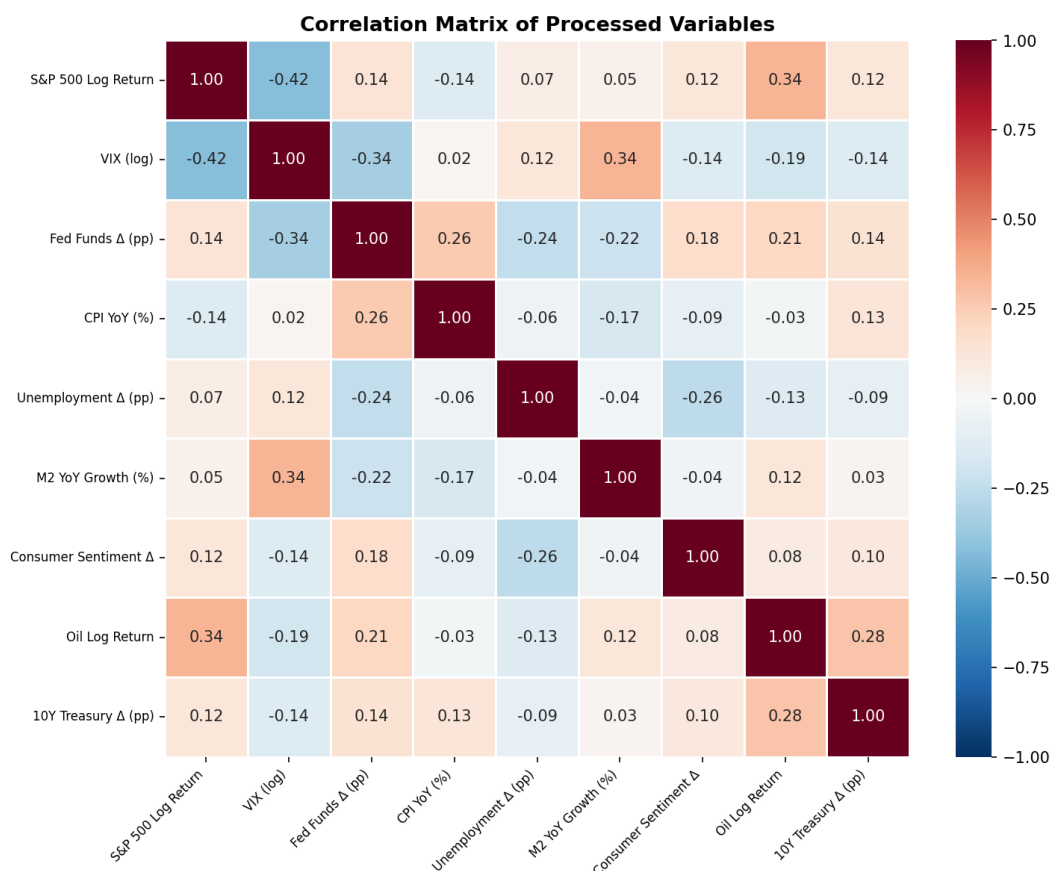


Figure 2: Pairwise correlation matrix of all processed variables.

Key correlations with S&P 500 return:

- **VIX:** -0.42 — strongest negative correlation (fear index)
- **Oil Return:** $+0.34$ — positive co-movement
- **Fed Funds Δ :** $+0.14$ — weak linear relationship

The modest correlations suggest simple linear relationships are insufficient, motivating our causal approach.

Rolling Correlation

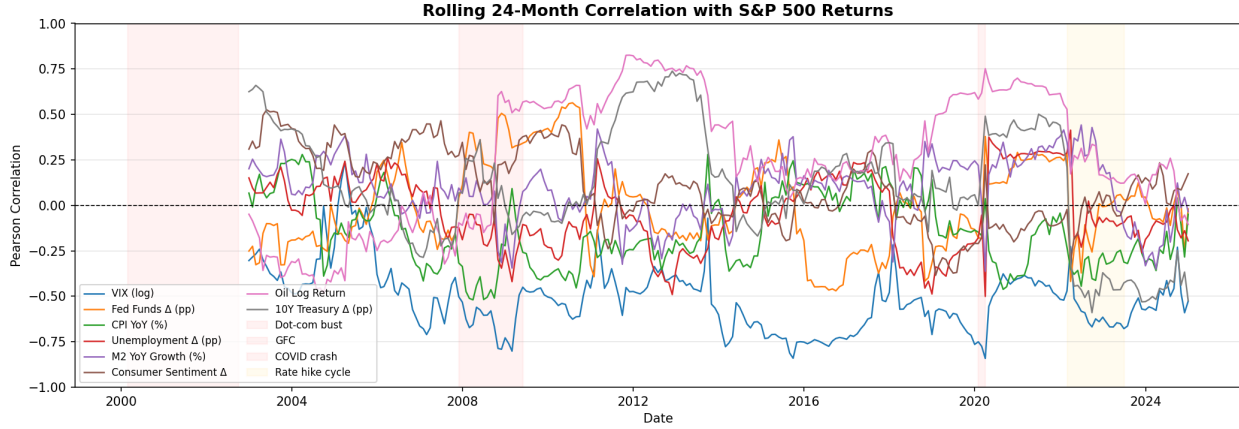


Figure 3: Rolling 24-month Pearson correlation of each variable with S&P 500 returns.

Correlations vary substantially over time — VIX–SP500 ranges from -0.9 to near 0 ; Oil–SP500 surged to $+0.8$ during 2008–2012. This instability directly motivates the use of CD-NOD to test for structural changes in causal mechanisms.

Methods & Results

Granger Causality

Granger causality tests whether past values of X help predict Y beyond Y ’s own history. We used a VAR model with AIC-selected lag of 2 months.

Table 2: Granger Causality: $X \rightarrow$ S&P 500 Return (lag = 2)

Variable	F-stat	p-value	Significant?
CPI YoY	3.698	0.026	Yes
Fed Funds Δ	1.417	0.244	No
Oil Return	1.250	0.288	No
VIX	0.077	0.926	No
Unemployment Δ	0.087	0.917	No
M2 Growth	0.285	0.752	No
Consumer Sentiment Δ	0.298	0.743	No
10Y Treasury Δ	0.471	0.625	No

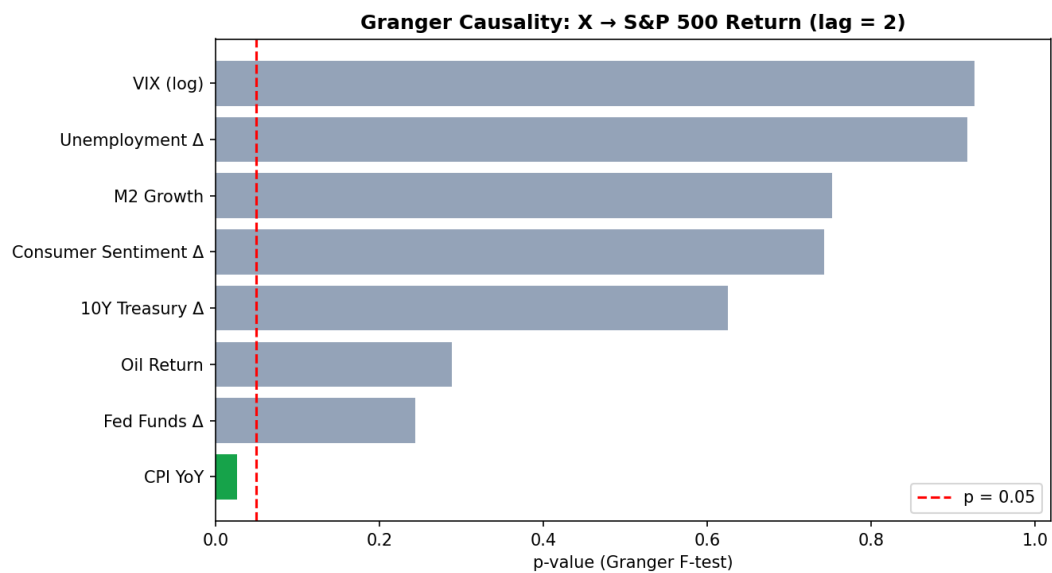


Figure 4: Granger causality p-values for $X \rightarrow$ S&P 500 Return. Only CPI YoY is significant (green bar, $p < 0.05$).

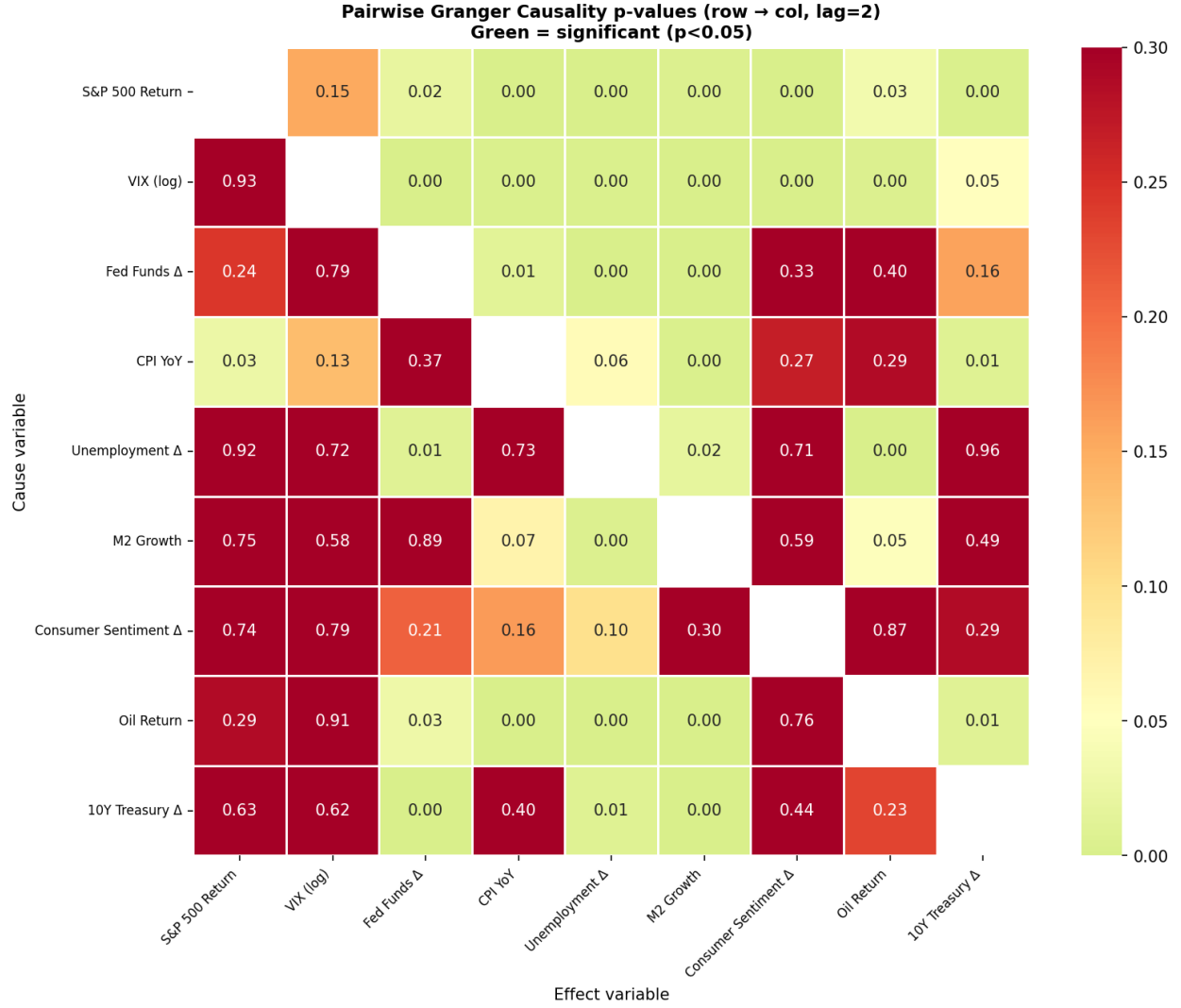


Figure 5: Full pairwise Granger causality matrix. Green cells indicate significant Granger causality ($p < 0.05$, lag = 2).

Only CPI significantly Granger-causes S&P 500 returns, consistent with market efficiency — most macroeconomic information is priced in quickly. The pairwise matrix reveals richer dynamics among macro variables: S&P 500 Granger-causes Fed Funds ($p = 0.02$); Oil Granger-causes CPI and unemployment.

PC Algorithm

The PC algorithm discovers causal structure from conditional independence tests (Fisher Z, $\alpha = 0.05$) on all 288 months of data.

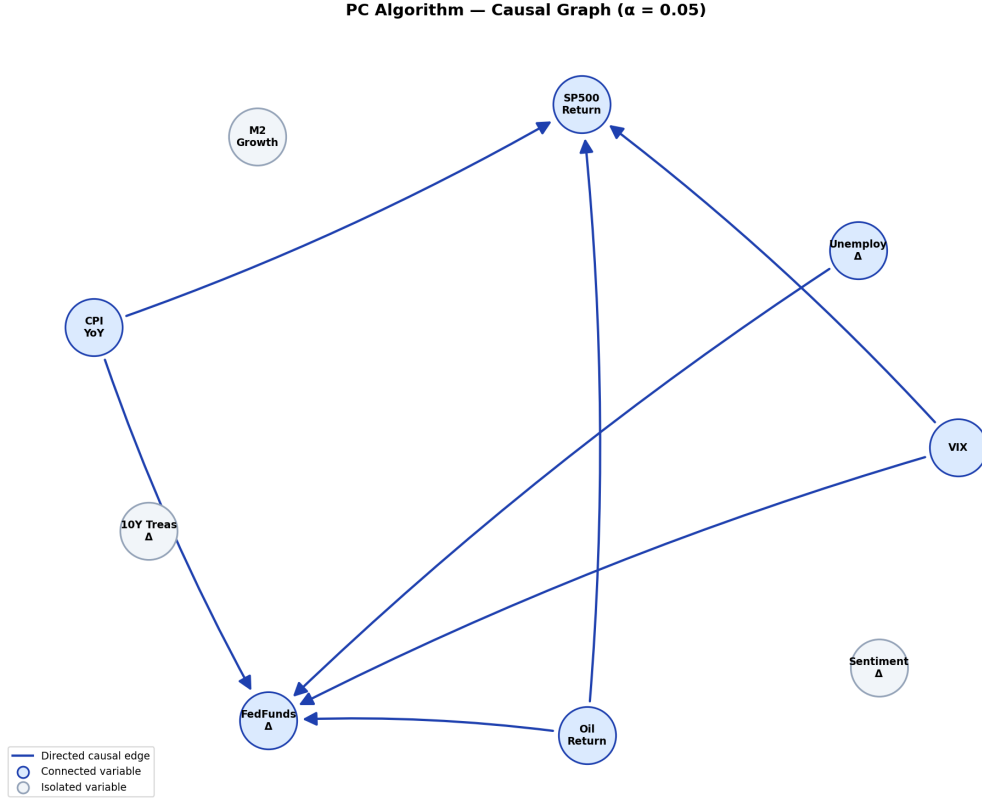


Figure 6: PC Algorithm causal graph ($\alpha = 0.05$). Blue nodes are causally connected; grey nodes (M2, Sentiment, 10Y Treasury) are isolated after conditioning.

Discovered edges:

- $VIX \rightarrow SP500 \text{ Return}$
- $CPI \text{ YoY} \rightarrow SP500 \text{ Return}$
- $Oil \text{ Return} \rightarrow SP500 \text{ Return}$
- $VIX \rightarrow Fed \text{ Funds } \Delta$
- $CPI \text{ YoY} \rightarrow Fed \text{ Funds } \Delta$
- $Unemployment \Delta \rightarrow Fed \text{ Funds } \Delta$
- $Oil \text{ Return} \rightarrow Fed \text{ Funds } \Delta$

Key observations: VIX and Oil have direct causal effects on SP500 that Granger missed, demonstrating PC captures conditional relationships that bivariate tests cannot. M2, Consumer Sentiment, and 10Y Treasury are isolated nodes — their information is subsumed by other variables after conditioning.

CD-NOD (Nonstationary Causal Discovery)

CD-NOD [1] extends PC to detect whether causal mechanisms change over time. We used a **continuous time index** (0 to 287) as the domain-change indicator, allowing detection of smooth structural shifts rather than imposing discrete regime boundaries.

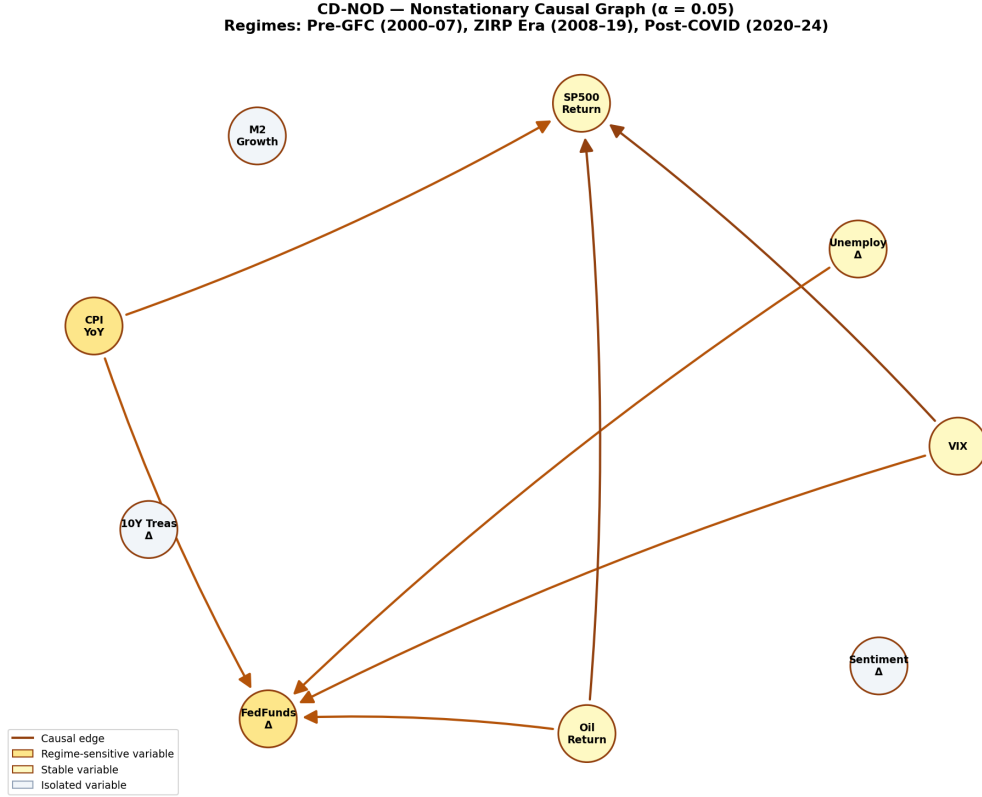


Figure 7: CD-NOD causal graph ($\alpha = 0.05$). Yellow nodes indicate regime-sensitive variables whose causal mechanisms changed significantly across the sample. Grey nodes are isolated.

Results:

- The overall causal graph structure is **identical to PC** (same 7 edges)
- **Two variables are regime-sensitive:** CPI YoY and Fed Funds Δ

This is economically meaningful: CPI’s mechanism changed across three eras (moderate pre-GFC inflation, near-zero ZIRP inflation, post-COVID surge); Fed Funds’ mechanism changed as the Fed’s reaction function shifted from conventional to unconventional policy and back to aggressive tightening. The structural stability of the remaining variables suggests the core causal architecture is robust across regimes.

Causal Effect Estimation

Using the PC-discovered graph and the **backdoor criterion**, we estimated causal effects via OLS regression with 1,000-iteration bootstrap confidence intervals.

Effects on S&P 500 Return

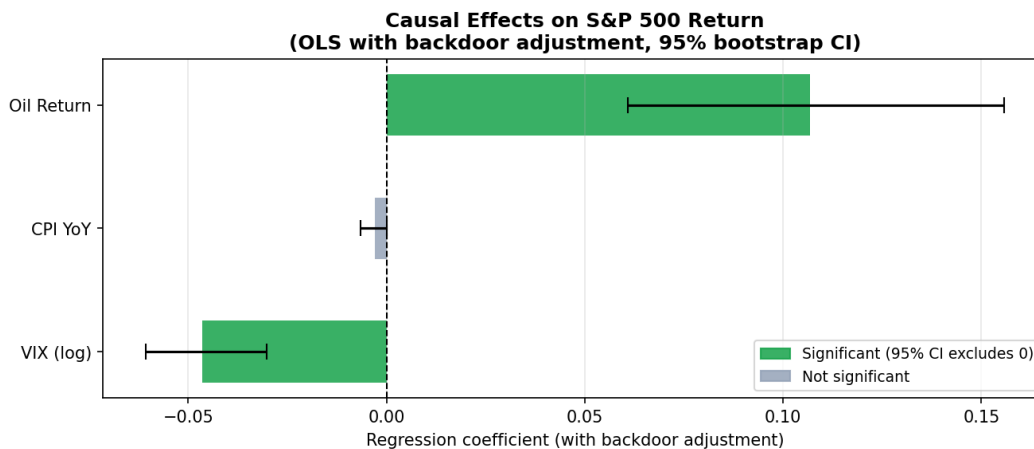


Figure 8: Causal effects on S&P 500 monthly log return. Error bars show 95% bootstrap CI. Green = significant (CI excludes 0).

Table 3: Causal Effects on S&P 500 Return (backdoor-adjusted OLS)

Cause	Coefficient	95% CI	Significant?
VIX (log)	-0.047	[-0.061, -0.030]	Yes
Oil Return	+0.107	[+0.061, +0.156]	Yes
CPI YoY	-0.003	[-0.007, +0.000]	No

CPI's near-zero direct effect suggests it affects the stock market *indirectly* through the Fed Funds channel, not directly.

Effects on Fed Funds Rate Δ

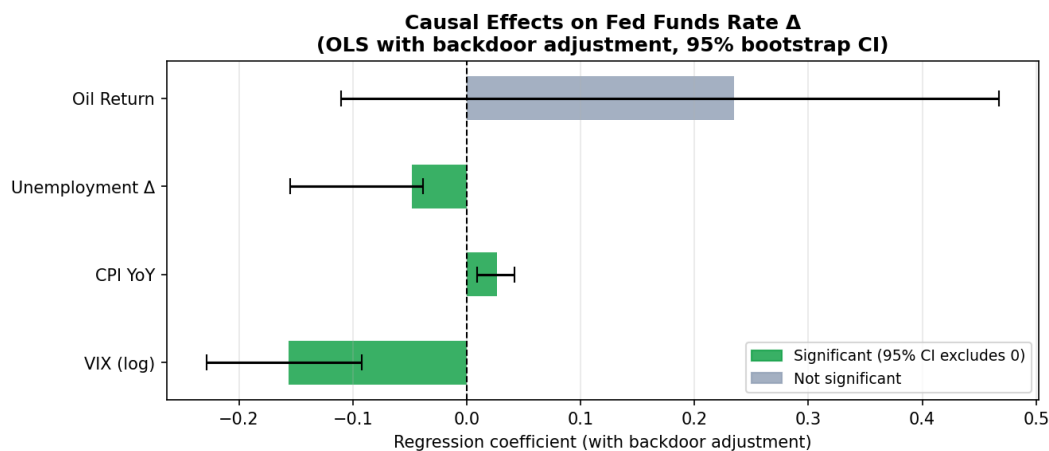


Figure 9: Causal effects on Fed Funds Rate monthly change. Error bars show 95% bootstrap CI.

Table 4: Causal Effects on Fed Funds Rate Δ (backdoor-adjusted OLS)

Cause	Coefficient	95% CI	Significant?
VIX (log)	-0.157	[-0.229, -0.092]	Yes
CPI YoY	+0.027	[+0.009, +0.042]	Yes
Unemployment Δ	-0.048	[-0.155, -0.039]	Yes
Oil Return	+0.235	[-0.111, +0.468]	No

Results are consistent with the Taylor Rule and the Fed's dual mandate: higher inflation raises rates; higher unemployment and market stress lower rates.

Summary of Findings

Table 5: Summary of Key Findings

Finding	Evidence
Most macro indicators do not linearly predict stock returns	Granger: only CPI significant
Causal structure is richer than linear prediction suggests	PC: 7 edges including VIX and Oil $\rightarrow$ SP500
Core macro-financial causal structure is stable over time	CD-NOD: same graph as PC
CPI and Fed Funds mechanisms changed across regimes	CD-NOD: 2 regime-sensitive variables
VIX and Oil are the strongest direct causes of stock returns	Effect estimation: both significant
CPI affects stocks indirectly via the Fed	CPI $\rightarrow$ SP500 near zero after backdoor adjustment

References

- [1] Huang, B., Zhang, K., Zhang, J., Ramsey, J., Sanchez-Romero, R., Glymour, C., & Schölkopf, B. (2020). *Causal discovery from heterogeneous/nonstationary data*. Journal of Machine Learning Research, 21(89), 1–53.
- [2] Spirtes, P., Glymour, C., & Scheines, R. (2000). *Causation, Prediction, and Search* (2nd ed.). MIT Press.
- [3] Granger, C. W. J. (1980). Testing for causality: A personal viewpoint. *Journal of Economic Dynamics and Control*, 2, 329–352.